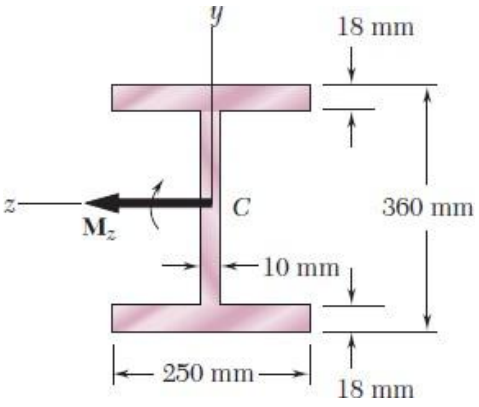
Question 1

The wide-flange beam shown is made of a high-strength, low alloy steel for which σ_Y = 345 MPa and σ_U = 450 MPa. Using a factor of safety of 3.0, determine the largest couple that can be applied to the beam when it is bent about the z axis. Neglect the effect of fillets.



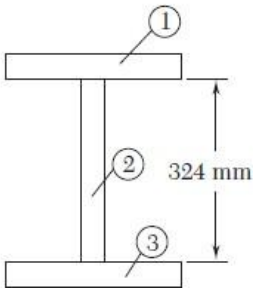
SOLUTION

$$I_1 = \frac{1}{12}bh^3 + Ad^2$$

$$= \frac{1}{12}(250)(18^3) + (250)(18)(171)^2$$

$$= 131.706 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12}(10)(324)^3 = 28.344 \times 10^6 \text{ mm}^4$$



$$I_3 = I_1 = 131.706 \times 10^6 \text{ mm}^4$$

$$I = I_1 + I_2 + I_3 = 291.76 \times 10^6 \text{ mm}^4 = 291.76 \times 10^{-6} \text{ m}^4$$

$$\sigma = \frac{Mc}{I} \qquad \text{where} \qquad c = \frac{360}{2} = 180 \text{ mm} = 0.180 \text{ m}$$

$$\sigma_{all} = \frac{\sigma_U}{F.S.} = \frac{450 \times 10^6}{3.0} = 150 \times 10^6 \text{ Pa}$$

$$M_{all} = \frac{\sigma_{all} I}{c} = \frac{(150 \times 10^6)(291.76 \times 10^{-6})}{0.180} = 243 \times 10^3 \text{ N} \cdot \text{m}$$

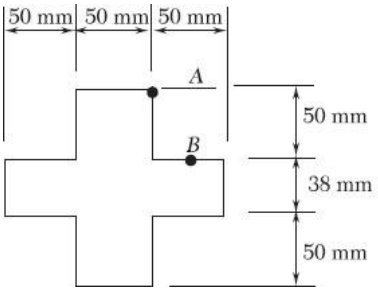
$$= 243 \text{ kN} \cdot \text{m}$$

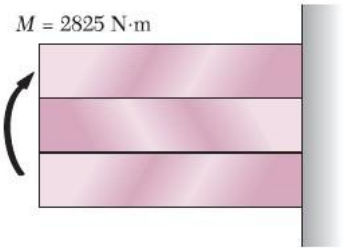
◄

Question 2

Will be answered in the Lecture-tute

Knowing that the couple shown acts in a vertical plane (about ahorizontal axis), determine the stress at point A, point B.

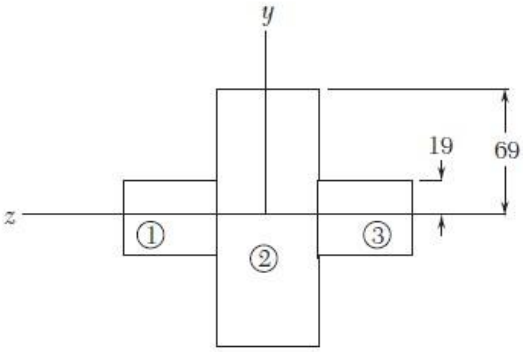




SOLUTION

For rectangle $I = \frac{1}{12} bh^3$

For cross sectional area



$$I = I_1 + I_2 + I_3 = \frac{1}{12}(50)(38)^3 + \frac{1}{12}(50)\,(138)^3 + \frac{1}{12}(50)(38)^3$$

$$= 11407\,566 \text{ mm}^4$$

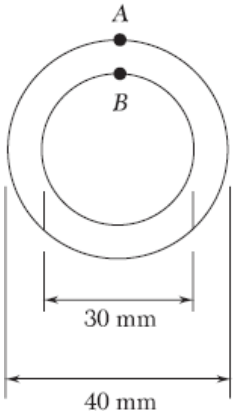
$$= 11.4 \times 10^{-6} \text{ m}^4$$

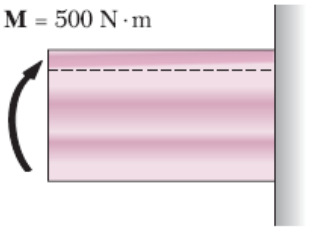
(a) $y_A = 69$ mm $\quad \sigma_A = -\frac{My_A}{I} = -\frac{(2825)(0.069)}{11.4 \times 10^{-6}} = -17.09$ MPa ◄

(b) $y_B = 19$ mm $\quad \sigma_B = -\frac{My_B}{I} = -\frac{(2825)(0.019)}{11.4 \times 10^{-6}} = -4.7$ MPa ◄

Question 3

Knowing that the couple shown acts in a vertical plane (about a horizontal axis), determine the stress at point A, point B.





SOLUTION

$$r_i = \frac{1}{2} d_i = 15 \text{ mm} \qquad r_0 = \frac{1}{2} d_0 = 20 \text{ mm}$$

$$I = \frac{\pi}{4}(r_o^4 - r_i^4) = \frac{\pi}{4}(20^4 - 15^4)$$

$$= 85.903 \times 10^3 \text{ mm}^4 = 85.903 \times 10^{-9} \text{ m}^4$$

(a) $y_A = 20 \text{ mm} = 0.020 \text{ m}$ $\quad \sigma_A = -\dfrac{My_A}{I} = -\dfrac{(500)(0.020)}{85.903 \times 10^{-9}}$

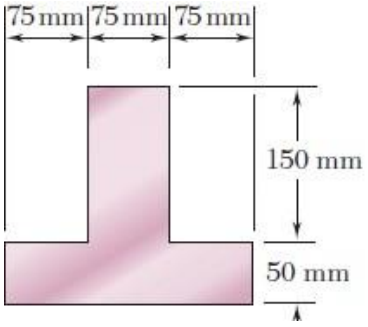
$$= -116.4 \times 10^6 \text{ Pa} = -116.4 \text{ MPa}$$ ◄

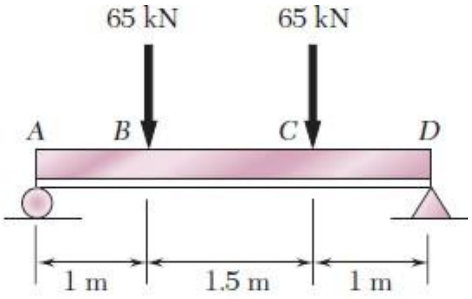
(b) $y_B = 15 \text{ mm} = 0.015 \text{ m}$ $\quad \sigma_B = -\dfrac{My_B}{I} = -\dfrac{(500)(0.015)}{85.903 \times 10^{-9}}$

$$= -87.3 \times 10^6 \text{ Pa} = -87.3 \text{ MPa}$$ ◄

Question 4

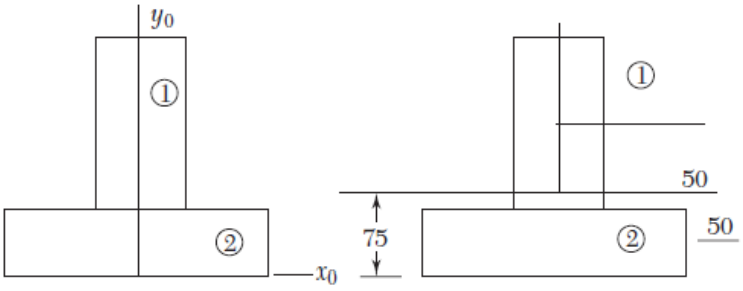
Two vertical forces are applied to a beam of the cross section shown. Determine the maximum tensile and compressive stresses in portion BC of the beam.





SOLUTION

	A, mm^2	$\bar{y}_0$, mm	$A\bar{y}_0$, mm^3
1	11250	125	1,406,250
2	11250	25	281,250
Σ	22500		1,687,500



$$\bar{Y}_0 = \frac{1{,}687{,}500}{22500} = 75 \text{ mm}$$

Neutral axis lies 75 mm above the base.

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12}(75)(150)^3 + (11250)(50)^2 = 49.2 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12}(225)(50)^3 + (11250)(50)^2 = 30.4 \times 10^6 \text{ mm}^4$$

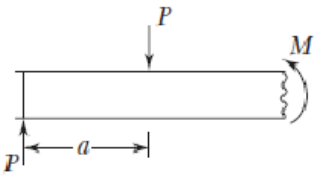
$$I = I_1 + I_2 = 49.2 \times 10^6 + 30.4 \times 10^6 = 79.6 \times 10^6 \text{ mm}^4 = 79.6 \times 10^{-6} \text{ m}^4$$

$$y_{top} = 125 \text{ mm} \qquad y_{bot} = -75 \text{ mm}$$

$$M - Pa = 0 \qquad M = Pa = (65 \times 10^3)(1.0) = 65 \text{ kN} \cdot \text{m}$$

$$\sigma_{top} = -\frac{My_{top}}{I} = -\frac{(65 \times 10^3)(0.125)}{79.6 \times 10^{-6}} = -102 \text{ MPa}$$

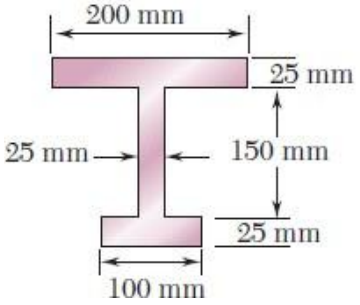
$$\sigma_{bot} = -\frac{My_{bot}}{I} = -\frac{(65 \times 10^3)(-0.75)}{79.6 \times 10^{-6}} = 61.2 \text{ MPa}$$

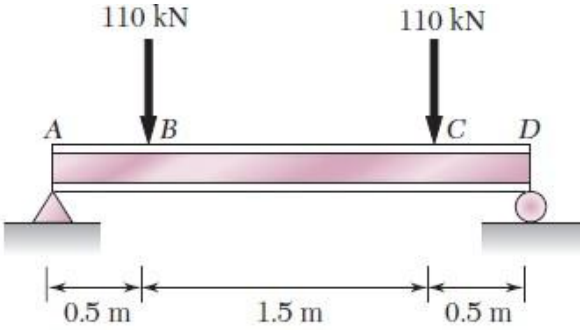


Question 5

Will be answered in the Lecture-tute

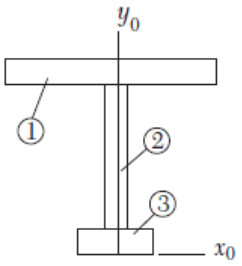
Two vertical forces are applied to a beam of the cross section shown. Determine the maximumtensile and compressive stresses in portion BC of the beam

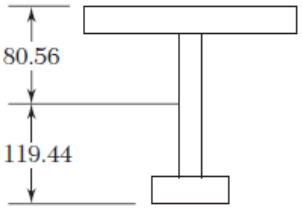




SOLUTION

	A, mm^2	$\bar{y}_0$, mm	$A\bar{y}_0$, mm^3
1	5000	187.5	937500
2	3750	10.0	375000
3	2500	12.5	31250
Σ	11250		1,343,750





$$\bar{Y} = \frac{1{,}343{,}750}{11{,}250} = 119.44 \text{ mm}$$

Neutral axis lies 119.44 mm above the base.

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12}(200)(25)^3 + (5000)(68.06)^2$$
$$= 23.42 \times 10^6 \text{ mm}^4 = 23.42 \times 10^{-6} \text{ m}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12}(25)(150)^3 + (3750)(19.44)^2$$
$$= 8.448 \times 10^6 \text{ mm}^4 = 8.448 \times 10^{-6} \text{ m}^4$$

$$I_3 = \frac{1}{12} b_3 h_3^3 + A_3 d_3^2 = \frac{1}{12}(100)(25)^3 + (2500)(106.94)^2 = 28.72 \times 10^6 \text{ mm}^4$$
$$= 28.72 \times 10^{-6} \text{ m}^4$$

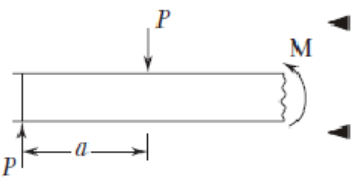
$$I = I_1 + I_2 + I_3 = (23.42 + 8.448 + 28.72)\ 10^{-6} = 60.588 \times 10^{-6} \text{ m}^4$$

$$y_{top} = 80.56 \text{ mm} \qquad y_{bot} = -119.44 \text{ mm}$$

$$M - Pa = 0 \qquad M = Pa = (110)(0.5) = 55 \text{ kN} \cdot \text{m}$$

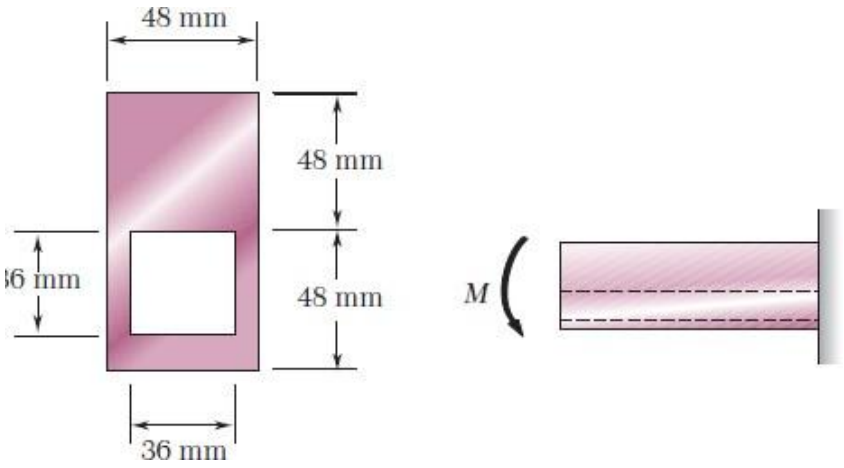
$$\sigma_{top} = -\frac{My_{top}}{I} = -\frac{(55 \times 10^3)(0.08056)}{60.588 \times 10^{-6}} = -73.13 \text{ MPa}$$

$$\sigma_{bot} = -\frac{My_{bot}}{I} = -\frac{(55 \times 10^3)(0.11944)}{60.588 \times 10^{-6}} = 108.42 \text{ MPa}$$



Question 6
Will be answered in the Lecture-tute

Knowing that for the extruded beam shown the allowable stress is 120 MPa in tension and 150 MPa in compression, determine the largest couple M that can be applied.



SOLUTION

	A, mm^2	$\bar{y}_0$, mm	$A\bar{y}_0$, mm^3
(1) solid rectangle	4608	48	221184
(2) square cutout	– 1296	30	– 38880
Σ	3312		182304

$$\overline{Y} = \frac{182304}{3312} = 55.04 \text{ mm}$$

Neutral axis lies 55.04 mm above bottom.

$$y_{top} = 96 - 55.04 = 40.96 \text{ mm} = 0.04096 \text{ m}$$

$$y_{bot} = -55.04 \text{ mm} = -0.05504 \text{ m}$$

$$I_1 = \frac{1}{12} b_1 h_1^3 + A_1 d_1^2 = \frac{1}{12} (48)(96)^3 + (48)(96)(7.04)^2 = 3.7673 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12} b_2 h_2^3 + A_2 d_2^2 = \frac{1}{12} (36)(36)^3 + (36)(36)(25.04)^2 = 0.9526 \times 10^6 \text{ mm}^4$$

$$I = I_1 - I_2 = 2.8147 \times 10^6 \text{ mm}^4 = 2.8147 \times 10^{-6} \text{ m}^4$$

$$|\sigma| = \left| \frac{My}{I} \right| \qquad \therefore \qquad M = + \left| \frac{\sigma I}{y} \right|$$

Top: tension side $$M = \frac{(120 \times 10^6)(2.8147 \times 10^{-6})}{0.04096} = 8.25 \times 10^3 \text{ N} \cdot \text{m}$$

Bottom: compression $$M = \frac{(150 \times 10^6)(2.8147 \times 10^6)}{0.05504} = 7.67 \times 10^3 \text{ N} \cdot \text{m}$$

M_{all} is the smaller value $$M = 7.67 \times 10^3 \text{ N} \cdot \text{m} = 7.67 \text{ kN} \cdot \text{m}$$ ◄